ASignX
Detailed progress report

21 October 2026 | System review, data preparation, training, investigation and reevaluation

Outcome: data traceability and validation improved substantially. A separate model was trained and evaluated, but it did not improve overall historical test performance. The retained model remains the recommended baseline; no replacement was promoted.

Milestone	Recorded result
New data prepared	4,995 filtered images; 4,118 accepted landmark samples
Candidate trained	50 epochs; best checkpoint selected at epoch 36
Source recovery	96,145 existing rows linked to a recovered import archive
Verification	114 tests passed; 13 existing export/tracing warnings
Retained test macro F1	59.49% unthresholded; 63.06% at confidence threshold 0.8
Candidate test macro F1	56.49% unthresholded; 60.98% at confidence threshold 0.8

What this report covers

The report consolidates the initial system review, interpretation of recognition errors, external Parquet filtering and extraction, isolated retraining, original-image recovery, source and split audits, code safeguards, and fresh offline evaluation. It also records what the results do not establish and the next data-collection work.

Existing capabilities documented in the 20 October repository review are identified as baseline context. Later work performed in this conversation is distinguished from that earlier implementation. Proposed collection targets are plans, not recordings already collected.

Scope of the evidence

The reported recognition scores are offline landmark classification results. They do not measure end-to-end webcam recognition, detector failure rates on a complete image corpus, continuous-stream false activations or new-signer reliability. The historical test has been inspected repeatedly during development.

Reading guide

Pages 2-5 cover architecture, preprocessing, evaluation meaning and the baseline. Pages 6-9 cover external data, training and complete performance comparisons. Pages 10-14 document source recovery, original images, deeper checks and software safeguards. Pages 15-16 list deliverables, commands and the next collection/evaluation plan.

Evidence: output/recognition-reevaluation/summary.json; output/weak-class-training/comparison.json; output/data-repair/verification.json

1. Progress and system architecture

Stage	Work and outcome
Initial review	Inspected source, model checkpoints, data models, training code, split manifests and saved metrics. Produced the first system report and PDF.
Data expansion	Filtered weak classes from a Parquet image dataset and extracted a separate landmark CSV with source-row traceability.
Model experiment	Prepared an isolated combined dataset, checked overlap and trained a 50-epoch candidate without replacing production files.
Data investigation	Recovered a hashed source archive and original images; checked source coverage, stored geometry and numerical near matches.
Safeguards and reevaluation	Added provenance and validation controls, passed 114 tests, reran static/dynamic inference and clarified review examples.

Current recognition flow

Webcam, image or video input passes through OpenCV and MediaPipe hand detection. Each hand yields 21 x/y/z landmarks and handedness. Static or sequence preprocessing prepares the geometry. Motion routing selects the static or dynamic classifier, followed by confidence filtering and temporal stabilization. The application displays labels or emits recognition events.

Baseline capabilities already present

The Python application supports up to two tracked hands, threaded capture, camera controls, static and dynamic recording tools, image/video import, CSV merging, training and evaluation. Timestamped live sampling, poor-window rejection, both-hand motion routing, calibration controls and TypeScript stabilization helpers were documented in the earlier review.

The retained static model covers 24 alphabet labels, A-I and K-Y. The dynamic checkpoint covers J and Z only. Example gesture names elsewhere in documentation do not mean those classes are pretrained. Facial expression, body-relative meaning, grammar and sentence translation are outside the demonstrated trained scope.

Evidence: main.py; asignx/camera.py; asignx/detector.py; ml/predict.py; ml/predict_sequence.py; docs/system-state-current.md

2. Models, data contracts and preprocessing

Property	Static checkpoint	Dynamic checkpoint
Input	63 features per pose	30 steps x 126 features
Feature enrichment	91 features	182 features per step
Backbone	Residual MLP, widths 128 and 64	128-wide projection, two residual temporal-convolution blocks, four-head attention
Output	Cosine prototype head; 24 labels	Mean pooling, 64-D embedding, cosine head; J/Z
Parameter count	63,681	296,641

Static preprocessing contract

MediaPipe produces 21 points. The system subtracts the wrist coordinates, divides by the maximum distance from the wrist, and flattens the result into 63 float values. The CSV stores these normalized values before canonical reflection. The canonical checkpoint reflects x for detector-labelled Left hands during training and prediction; Right coordinates remain unchanged.

Inside the network, enrichment adds 15 joint-angle cosines, 10 fingertip distances and a three-component palm normal. These geometric features help the network represent articulation, but cannot recover detail that the detector did not locate correctly.

Data models and storage

Runtime structures include Landmark, HandLandmarks and HandDetectionResult. A static Sample contains label, session_id, handedness and features. CSV columns are label, session_id, handedness, preprocessing, then x0/y0/z0 through x20/y20/z20. SequenceSample stores a time-by-feature array plus metadata in compressed .npz files. Two-hand sequences use Left then Right slots.

Dynamic preprocessing uses a clip-wide reference and scale, preserves wrist travel, interpolates missing detections and resamples complete clips. Its rolling live windows remain different from complete-gesture training examples. The class name SignLSTM wraps the retained tcn_attention backbone; it does not imply the saved network is an LSTM.

Dataset-format questions resolved

Grayscale images can be imported, but their value depends on landmark detection quality; the sign classifier receives geometry rather than image colours. JSON, CSV, NumPy and Parquet compatibility depends on contents and coordinate contracts, not extensions. Two-dimensional points or body joints are not direct replacements for 21 x/y/z hand landmarks. The actual completed conversion used Parquet image bytes; no separate JSON conversion was performed.

Evidence: ml/data_io.py; ml/sequence_io.py; ml/preprocessing.py; ml/model.py; ml/sequence_model.py

3. Baseline data and evaluation meaning

Partition	Original static samples	Purpose
Training	109,223	Update weights; learn sign patterns
Validation	13,791	Choose checkpoints and training settings
Test	36,973	Evaluate the selected model on reserved rows
Total	159,987	24 labels; 1,639 session strings

The original dataset contains 98,909 Right and 61,078 Left samples. Class sizes range from N = 5,281 to L = 8,252, a moderate 1.56x imbalance. Session-based grouping, rather than a direct random row split, explains why the row fractions differ from nominal 70/15/15 targets.

What test macro F1 means

The test score uses only rows assigned to test session IDs. F1 balances precision and recall for each class; macro F1 is the equal average of those 24 class F1 scores. It is not a percentage of correct predictions, not an average over sessions, and not a combination of training, validation and test scores. Accuracy is reported separately.

Why three partitions matter

Training is study material, validation is a practice exam used to choose a checkpoint, and test is a reserved exam. Evaluating only training examples can reward memorization. A high validation score can still coexist with a low test score when source, viewpoint or acquisition conditions differ.

Identity limitation

Session IDs do not consistently identify independent people or recordings. Some represent an entire source for one letter; older IDs include artificial row chunks. Keeping session strings separate therefore does not establish signer independence across all labels. Real global signer/source groups are needed for stronger claims.

Adding test samples changes the measurement, not the model. Adding representative training samples can help learning after retraining. A new test set should be reserved before development and should not be selected because the current model already recognizes its samples.

Evidence: ml/models/sign_classifier_canonical.splits.json; ml/dataset.py; ml/benchmark.py; ml/evaluate.py

4. Baseline findings and earlier experiments

Metric	Retained static model
Validation accuracy	97.64%
Validation macro F1	97.33%
Test accuracy	53.19%
Test macro F1	59.49%
Validation/test accuracy gap	44.46 percentage points

The baseline audit localized major failures to imported groups. Its archive5 source-hint rows achieved 99.93% test accuracy, compared with 37.82% on hashed-import rows. These are diagnostic group strings, not verified signer identities. Later source recovery improved traceability for many hashed rows.

True sign to prediction	Retained test errors
K to V	3,304
M to N	2,437
Q to P	2,280
G to L	1,870
G to H	1,711

Earlier controlled screen documented on 20 October

Five paired 12-epoch variants used a fixed seed, batch size 512 and validation-only selection. They varied class weighting, augmentation or smoothing one setting at a time. The selected no-smoothing variant reached 97.31% validation macro F1 but only 54.98% test macro F1, below the retained 59.49%. No variant was promoted.

This short screen was not an equal-compute comparison with the retained 50-epoch, batch-64 run. It showed that those changes did not solve the problem in that experiment; it did not establish an optimal architecture or training recipe.

Low test F1 is evidence of a recognition limitation. It does not, by itself, prove a wrong label or faulty preprocessing. Removing difficult valid examples would weaken the benchmark rather than fix the model.

Evidence: output/f1-review/baseline/summary.json; output/f1-review/experiments/comparison.json; output/f1-review/release-decision.json

5. External Parquet data preparation

The supplied train-00000-of-00001.parquet contained 10,873 records with embedded image bytes and integer A-Z labels. Its metadata supplied the label-ID mapping. It was an image container, not a table of ready-made hand landmarks.

We selected classes with retained-model historical test F1 below 60%: C, D, G, H, K, L, M, N, P, Q, R and V. This retained 4,995 images and excluded 5,878. Original label IDs and embedded image content were preserved; the original file was not changed. The rule was a diagnostic prioritization choice, not a new independent benchmark design.

Class	Filtered images	Accepted landmarks
C	420	353
D	420	344
G	420	338
H	419	353
K	420	350
L	420	357
M	420	354
N	417	352
P	419	331
Q	380	294
R	420	347
V	420	345

Extraction accepted 4,118 samples, or 82.44% of selected images. It skipped 875 images with no detected hand and two with multiple hands. No statistical unusual-pose rejection or optional similarity pruning was applied. The extraction report recorded no other rejection categories for this run.

Each image was saved with its original Parquet row number. Image hashes, source-row mappings and landmark-row provenance were preserved. Original images can now be inspected when an extracted row is ambiguous. The source schema did not provide signer identities.

Diagnostic inference before retraining

The retained model scored 66.76% accuracy on the 4,118 accepted samples. Macro F1 over the 12 present target classes was 68.71%; M F1 was 74.22% and N F1 62.25%. These figures are not directly comparable with the historical 24-class test macro F1. The images lacked verified signer independence and later became candidate training data, so this is not a reserved final test.

Evidence: output/datasets/weak-classes/manifest.json; extraction-summary.json; image-provenance.csv; landmark-provenance.csv

6. Candidate training design and execution

Item	Recorded configuration
Training data	Original 159,987 rows plus 4,118 extracted rows
Combined CSV	164,105 rows, saved separately
Training partition	113,341 rows; all new data training-only
Validation/test	Original 13,791 / 36,973 rows and indices unchanged
Initialization and architecture	From scratch; canonical 24-class residual MLP
Schedule	50 epochs; batch size 64; seed 42; CPU
Optimizer	AdamW; initial learning rate 0.002; weight decay 0.0001
Loss and regularization	Inverse-frequency weights; smoothing 0.05; dropout 0.2; rotation/jitter augmentation
Checkpoint selection	Validation macro F1, with validation loss tie-breaker
Selected checkpoint	Epoch 36; validation macro F1 97.85%

Overlap screening

Before combination, all 4,118 new samples were compared with all original rows in canonical landmark space. A conservative L2 distance cutoff of 0.015 excluded close matches across all original partitions and labels. No new samples fell within that cutoff, so all were retained for training. The CSV overlap review records each nearest original row.

A distance check cannot establish different signers, detect every transformed duplicate or prove independent sources. Consequently, the run is labelled a diagnostic experiment rather than a verified new-signer benchmark. The original CSV and retained checkpoints were protected by SHA-256 checks.

Release decision

Test inference occurred after checkpoint selection. The candidate improved M and N but reduced overall test accuracy and macro F1. It was saved for inspection and explicit launch only. No default launcher or retained checkpoint was replaced. A single from-scratch run cannot isolate added data as the sole cause of any gain or regression.

Evidence: output/weak-class-training/preparation.json; overlap-review.csv; model.metrics.json; model.splits.json; release-decision.json

7. Complete static test F1 comparison

Class	Retained F1	Candidate F1	Change, points
A	74.52%	76.62%	+2.10
B	95.56%	92.94%	-2.62
C	42.98%	55.15%	+12.16
D	32.88%	20.69%	-12.19
E	75.91%	85.06%	+9.15
F	92.25%	91.17%	-1.09
G	21.71%	21.64%	-0.07
H	36.29%	35.87%	-0.42
I	96.52%	96.65%	+0.13
K	45.69%	46.79%	+1.10
L	36.83%	42.38%	+5.55
M	36.73%	46.72%	+9.99
N	22.81%	24.85%	+2.04
O	72.03%	35.58%	-36.45
P	18.56%	19.20%	+0.64
Q	50.40%	48.37%	-2.03
R	56.22%	37.32%	-18.90
S	76.25%	49.55%	-26.71
T	70.65%	72.48%	+1.83
U	87.74%	90.97%	+3.23
V	23.44%	23.51%	+0.07
W	99.09%	98.55%	-0.54
X	68.28%	53.22%	-15.05
Y	94.42%	90.59%	-3.83

Overall macro F1: retained 59.49%; candidate 56.49%, a decline of 3.00 percentage points. **Accuracy:** retained 53.19%; candidate 48.01%. All figures use the same historical 36,973 test rows without confidence rejection.

M improved by about 9.99 points and N by 2.04, but regressions elsewhere outweighed those gains. O, R, S and X are examples of substantial candidate declines. Improving selected weak classes is insufficient if the complete vocabulary becomes less reliable.

Evidence: output/recognition-reevaluation/summary.json; retained-test-predictions.csv; candidate-test-predictions.csv

8. Fresh reevaluation and confidence filtering

After source recovery and validation safeguards, we reran inference using both fixed checkpoints and their matching manifests. Candidate validation/test indices were checked against the original indices, and the combined CSV was verified to preserve the original data prefix. No training, threshold fitting or data relabeling occurred during reevaluation.

Metric	Retained	Candidate
Validation macro F1	97.33%	97.85%
Test accuracy	53.19%	48.01%
Test macro F1, no rejection	59.49%	56.49%
Test macro F1, threshold 0.8	63.06%	60.98%
Accepted coverage at 0.8	85.79%	85.13%
Accepted-only accuracy at 0.8	56.81%	53.60%
Rejected known test samples	5,253	5,497

At confidence threshold 0.8, rejected known signs remain false negatives. The thresholded F1 increase is therefore not an accepted-only calculation. Accepted-only accuracy answers a different question and excludes rejected samples; it must always be read with coverage. Neither model has representative unknown-gesture validation.

Dynamic J/Z reevaluation

The recovered dynamic baseline contains 11 clips: seven training, two validation and two test. Fresh evaluation again returned 100% test macro F1 on one J and one Z clip. Reversal, frame shuffling, last-frame repetition and removal of wrist travel also retained perfect scores on those two clips. This does not establish movement understanding.

The earlier recovery found 16 portable-folder clips in total: 11 J and five Z. An exact 11-clip subset reproduced the saved baseline hash; the five additional clips were all J and did not provide extra Z or negative coverage. Signer identities remained unknown.

No end-to-end camera or continuous-stream recognition benchmark was run. Detector failures are outside these landmark classification scores. Timestamped sampling and regression tests are useful software checks, but are not substitutes for a labelled live-stream evaluation.

Evidence: output/recognition-reevaluation/summary.json; dynamic.json; report.md; ml/audit_sequence.py

9. Source recovery and the coverage gap

The original M-to-N error used session ext_c72cf97fc7fa5995335a73a1 and initially appeared to have unavailable source identity. Recomputing the historical import-session hash against available archives matched this session to Downloads/archive (4).zip. The same procedure recovered 24 sessions covering 96,145 rows, or about 60.10% of the original static dataset.

Partition	Recovered-source rows	Letters from that source
Training	64,409	A, B, C, D, E, F, H, L, N, P, R, S, T, U, V, Y
Validation	4,165	W
Test	27,571	G, I, K, M, O, Q, X

G, I, K, M, O, Q and X have test examples from this archive but no training examples of those letters from that archive. Other letters from the same archive are in training. This is a concrete source-by-class coverage difference and an explanation to investigate for the validation/test gap; it does not prove a single cause.

What was recovered and what remains unknown

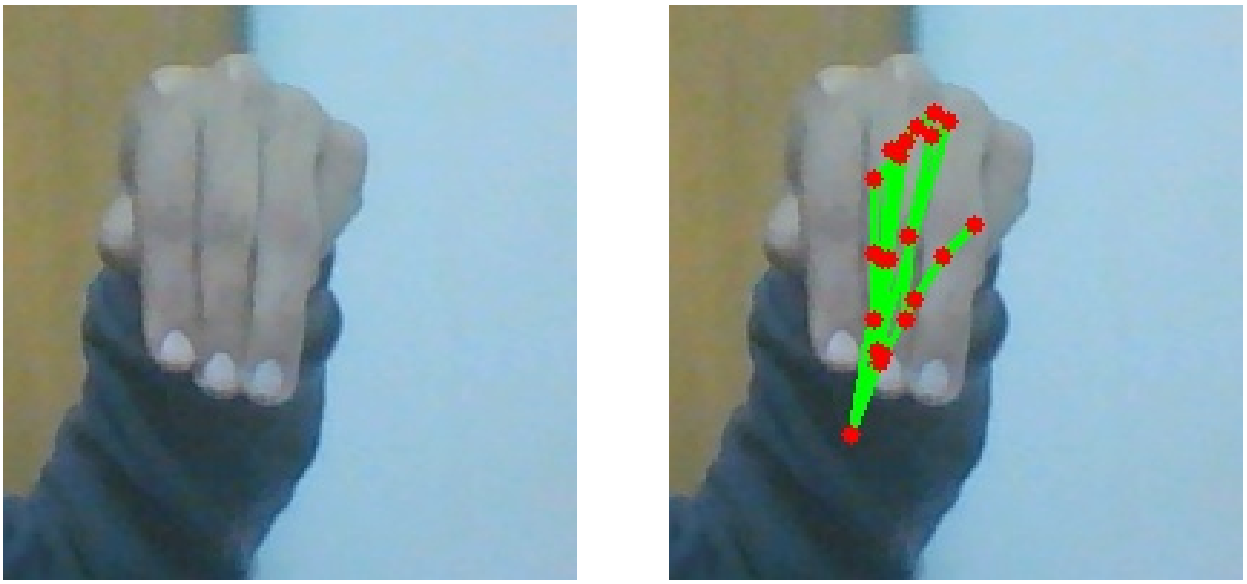
The exact hash match recovers the source tag used by the importer. A present-day content hash of the archive is recorded. It does not establish the image for every row, signer identities, independent recording visits or that the archive never changed. Legacy source strings and other unmatched identities remain unresolved.

The unmatched portion is 63,842 rows: 44,814 training, 9,626 validation and 9,402 test. A benchmark-plan template places the recovered archive into one global source group across labels and leaves unknown identities and split choices blank. It is intentionally incomplete and not ready for training.

Historical test rows were not moved into training to improve the score. Any future source/signer benchmark must have a separately documented partition plan and fresh model training; it cannot be presented as directly interchangeable with this historical benchmark.

Evidence: output/data-repair/quality-report.json; session-provenance.csv; benchmark-plan-template.csv; ml/data_quality.py

10. Recovered M-to-N error: row 110182



Left: matching original image. Right: detected landmark overlay. Original image resolution is 200 x 200 pixels; it is enlarged here for inspection.

Property	Verified observation
Stored label / partition	M / test
Stored and detected handedness	Right / Right
Current model output	N, confidence 96.16%
Stored M probability	Approximately 0.365%
Maximum coordinate difference	7.45e-9 after re-extraction
Archive member	ASL_Alphabet_Dataset/asl_alphabet_train/M/M (100).jpg

The source image reproduces every stored coordinate within approximately 0.0000000075 and agrees on handedness. This rules out a numerical extraction mismatch for this example. It does not prove that MediaPipe placed every point anatomically correctly or that the M label is linguistically correct.

The image has overlapping/partly hidden fingers and limited resolution. Those conditions can make thumb and folded-finger placement difficult to verify. Numerical consistency can coexist with uncertain detected geometry. Model confidence cannot resolve that uncertainty.

Decision: keep the M label unchanged and mark the discrepancy unresolved pending competent review of the original sign and geometry. Do not change M to N merely because the model is confident.

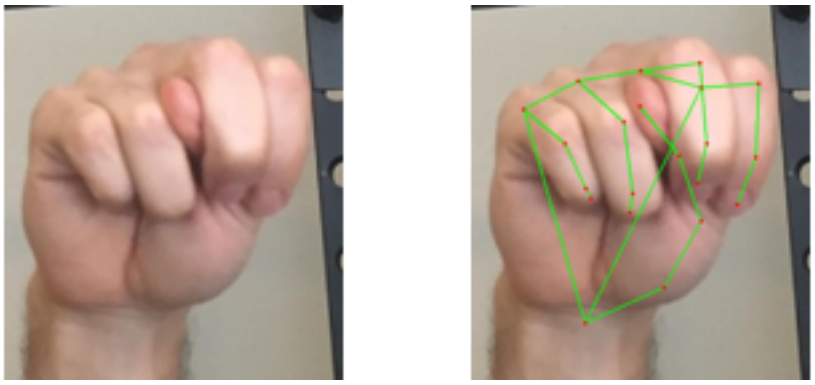
Evidence: output/data-recheck/row-110182-recovery.json; output/data-recheck/recheck.json; ml/recover_image.py

11. N, K and Q examples clarified

These rows were selected as source-image examples, not confirmed bad labels. The review originally did not make that distinction clear. We checked current predictions and partitions, then updated the review cards with explicit status. Their labels were not changed.

CSV row	Stored / predicted	Confidence	Partition
113222	N / N	95.72%	Training
100747	K / K	97.68%	Test
123354	Q / Q	93.04%	Test

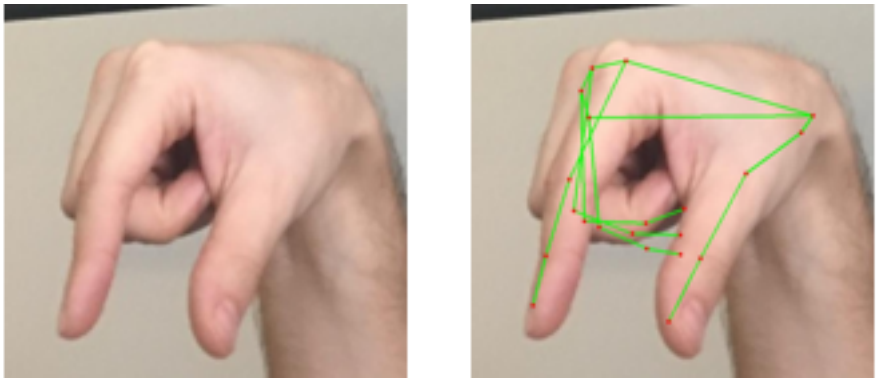
N
row 113222



K
row 100747



Q
row 123354



Hand-detail crops: original at left, overlay at right. The three source images are 1920 x 1920; crops improve legibility. Original files are preserved. Matching predictions do not independently verify labels. The N example was in training and is not evidence of held-out generalization.

Evidence: output/data-recheck/three-row-check.json; row-113222-recovery.json; row-100747-recovery.json; row-123354-recovery.json

12. Deeper data recheck

We recounted and validated the current original CSV after the investigation. It still contains 159,987 rows, 24 labels and 1,639 sessions, with its original SHA-256. Numerical shape, finite values, wrist-centering, scale, canonical split checks and exact contradictory-label checks passed. No rows or labels were changed.

Sampled evaluation rows	Validation	Test
Rows checked	2,400	2,400
Sampling rule	Up to 100 per class	Up to 100 per class
Closest-training cutoff	L2 <= 0.015	L2 <= 0.015
Matches within cutoff	0	0
Minimum observed distance	0.015001	0.015114
Median nearest-training distance	0.138057	0.193145

Each sampled row was compared with every original training row in canonical landmark space using seed 42. This is a class-balanced diagnostic sample, not an exhaustive check of every validation/test row. The cutoff matches the conservative similarity screen used in the candidate preparation.

The sampled test geometry is farther from training geometry on this measure. That supports a distribution difference worth investigating; it does not establish label errors or a unique cause of low F1. No matches below the cutoff is not proof that all images or signers are independent. Similar landmarks can also legitimately represent different signs.

Image and landmark interpretation

The four recovered originals reproduce stored geometry, but have different resolutions, hand scales, backgrounds and viewpoints. Correct serialization and normalization are narrower claims than correct anatomical landmark placement. Hidden thumb/finger details require image inspection and competent sign review.

Limits of the diagnosis

No confirmed preprocessing corruption, exact contradictory-label group or numerical defect was identified in the current original data. Possible label inconsistencies and signer/source dependence remain unverified. We did not manufacture a clean subset by deleting model errors, and no performance gain is claimed from the recheck.

Evidence: output/data-recheck/recheck.json; nearest-training-samples.csv; review.html; tmp/recheck_static_data.py

13. Software safeguards and verification

Change made	Purpose
Shared import-session hash helper	Keep import and source recovery on the same historical ID contract
Accepted-row import provenance	Bind each new output CSV row to image hash, original file path or ZIP member, session and handedness
Stored-feature and dataset hashes	Bind audit evidence to the exact stored numeric precision and committed CSV
Conflicting-label validation	Reject identical canonical features with different labels, including contradictions within training
Static preprocessing in manifests	Persist the handedness contract and reject incompatible reuse; retain historical compatibility
Source and image recovery tools	Report source coverage gaps and find geometrically matching originals without changing labels
Review-card clarification	Show prediction agreement/mismatch, confidence and partition explicitly

Latest recorded software checks

The full suite passed 114 tests after these changes, compared with the earlier 108-test snapshot. Thirteen existing export/tracing warnings remained. Tests covered source-hash recovery, source/class gap reporting, canonical contradictions, preprocessing-contract persistence, strict image matching and ZIP provenance surviving temporary extraction cleanup, alongside existing model, detector, training and live-replay checks.

The recovery CLI completed successfully. Original-image hashes were checked, dataset/source evidence was bound to the current dataset hash, and SHA-256 checks confirmed the original CSV and retained static/dynamic checkpoints were unchanged. The candidate checkpoint was also unchanged during fresh reevaluation.

The PDF build rechecks the relevant data/checkpoint hashes against the reevaluation evidence. It does not rerun training or the entire test suite. Passing software tests establishes implementation checks, not accurate recognition for new signers.

Evidence: output/data-repair/verification.json; tests/test_data_quality.py; tests/test_extractor_integrity.py; ml/data_io.py; ml/dataset.py; ml/train.py

14. Deliverables and reproducible workflow

Location, relative to repository	Contents
output/pdf/	Initial system report PDF and this detailed progress report
output/datasets/weak-classes/	Filtered Parquet, original images, extracted CSV, predictions, evaluation and row-level provenance
output/weak-class-training/	Combined CSV, overlap review, candidate checkpoint, frozen manifests, metrics, comparison and release decision
output/data-repair/	Recovered source mapping, coverage report, benchmark-plan template and row-110182 recovery
output/data-recheck/	Four recovered source-image examples, nearest-training diagnostic and clarified visual review
output/recognition-reevaluation/	Fresh static/dynamic metrics, per-row predictions and readable comparison report
docs/data-repair.md; docs/f1-workflow.md	Investigation details and evaluation/training guidance

Launch models explicitly from the project root

Retained model:

```
py -3.11 main.py --sign-model
ml/models/sign_classifier_canonical.pth --exposure auto
```

Candidate, for inspection only:

```
py -3.11 main.py --sign-model
output/weak-class-training/model.pth --exposure auto
```

The displayed commands wrap for readability; enter each launch command on one terminal line. A previous "no such file or directory" report was addressed by checking that main.py and the candidate existed, then running from the project root. No application path bug was established.

Reproduce checks

```
py -3.11 -m pytest -q
py -3.11 -m ml.data_quality --source
"C:\Users\jibbi\Downloads\archive (4).zip"
```

The reevaluation orchestration script is tmp/reevaluate_recognition.py; it writes to output/recognition-reevaluation/. The 50-epoch training orchestration is tmp/train_weak_candidate.py and deliberately refuses to overwrite existing candidate outputs. Preserve its experiment artifacts rather than rerunning into the same directory.

Evidence: Existing artifact directories and the listed scripts; all paths inspected during report preparation

15. Next steps and acceptance criteria

1. Verify labels and acquisition conventions

Review original images for M/N, K/V, Q/P and G/H/L with someone familiar with the target sign alphabet. Check thumb/finger geometry, mirroring, detector handedness and source conventions. Record evidence-backed corrections in a versioned log. Keep unresolved cases unresolved; a confident model prediction is not a replacement label.

2. Establish a trustworthy independent benchmark

Collect all 24 static labels from 5-10 new signers, with 5-10 independent attempts per class per signer, ideally spread over two visits. This is a starting target of 600-2,400 attempts, not a statistical guarantee. Keep each signer entirely in one partition across all labels. Preserve original images/video, pseudonymous signer IDs, recording IDs, device and conditions.

Use normal lighting and comfortable variation in viewing angle, distance and background. Include both handedness conditions under a consistent capture convention. Do not count hundreds of adjacent frames from one hold as hundreds of independent attempts. Do not select final test images because the model already gets them right.

3. Separate learning from final measurement

Reserve final test signers before training. Use grouped validation among the remaining data for settings and checkpoint selection. Because the historical test has already informed development, repeated evaluation on it provides comparison evidence but not a fresh independent final test. More test samples can raise or lower the measured score without improving the model.

4. Measure realistic live behaviour and dynamic motion

Collect separate idle, transition, incorrect-path, matching-handshape hold and repeated-sign streams. Measure event precision/recall, false emissions per minute, missed gestures and decision delay. Expand J/Z across people, speeds, visits and devices before considering a larger vocabulary. Include tracking failures in end-to-end reporting.

5. Promote only a justified replacement

Require an overall macro-F1 improvement on a prespecified representative benchmark, review per-class regressions, report uncertainty by independent signer/recording, and verify live behaviour. Profile capture, detection, preprocessing, inference and rendering separately. Verify ONNX export and Python/browser output parity before claiming browser deployment complete.

Current decision: keep the retained canonical model and confidence threshold 0.8. Infrastructure and evidence improved; overall recognition performance has not yet improved. New verified data and independent evaluation are the next substantive work.